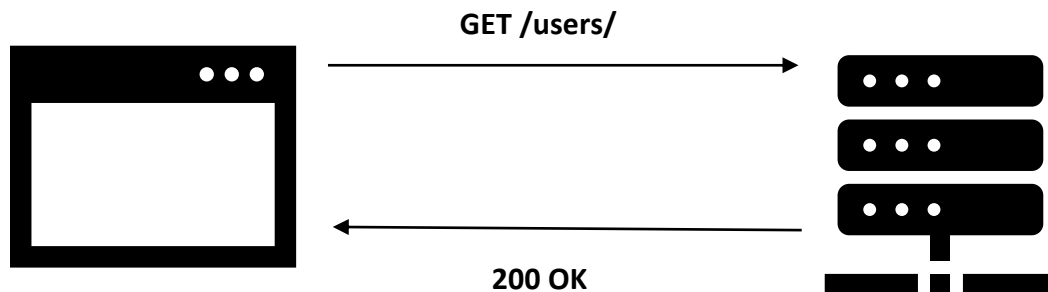


HTTP Status Codes

An HTTP status code is a 3-digit number a server sends in response to a client's request. It tells the client what happened to the request.



Why Are Status Codes Important?

Status codes matter because they give the client and server a standard way to communicate a request's result. The client can use the status code to decide what to do next.

Example

```
if (response.status === 200) {  
  // Display the data  
}
```

```
if (response.status === 404) {  
  // Show "User not found"  
}
```

```
if (response.status === 401) {  
  // Ask user to log in  
}
```

They are especially important for:

- Frontend applications such as React, Next.js, Angular, etc.
- REST APIs
- Backend development
- Authentication and authorization
- Error handling
- Debugging
- Web browsers and search engines
- Communication between different services

Important HTTP Status Codes

Status Code	Status	Meaning	Common Use Case
200	OK	Request was successful	Successfully fetching user/product data
201	Created	A new resource was successfully created	Creating a new user/order
400	Bad Request	Server cannot process the request because the request is invalid	Invalid JSON, missing required data
401	Unauthorized	Authentication is required or authentication failed	User is not logged in / invalid token
403	Forbidden	Server understood the request but refuses to allow it	User is logged in but doesn't have permission
404	Not Found	Requested resource does not exist	User/product/page doesn't exist
500	Internal Server Error	Unexpected error occurred on the server	Database/server-side error
204	No Content	Request succeeded but there is no response body	Successful DELETE operation
405	Method Not Allowed	HTTP method is not supported for that resource	Sending POST to a GET-only endpoint
409	Conflict	Request conflicts with the current state of the resource	Duplicate username/email
429	Too Many Requests	Client has sent too many requests	API rate limiting
503	Service Unavailable	Server is temporarily unable to handle the request	Server overload/maintenance

Understanding the Status Code Categories

Range	Category	Meaning
1xx	Informational	Request is being processed
2xx	Success	Request was successfully processed
3xx	Redirection	Client needs to take another action/location
4xx	Client Error	Something is wrong with the client's request
5xx	Server Error	Server failed while processing a valid request

How to set a status code in Node.js?

```
const http = require("http");
const server = http.createServer((req, res) => {
  res.statusCode = 200;
  res.end("User data fetched successfully");
});
server.listen(3000, ()=>{
  console.log("Server is running...")
});
```